TRex tracking

What this is about

The lab is testing whether daytime bumblebees can find their way under low light using polarised light cues. I had 84 videos to work with, each one showing a single bumblebee walking into a round arena, finding a feeder in the middle, staying there for a while, and then walking back out.

My job was to get TRex to follow the bee in every video and save the (x, y) position in every frame. After that, the trajectories will be used to compute behavioural metrics (speed, exit angle from the inner circle, drift, etc.). I have done the TRex part; the analysis step is what you will pick up.

Below is everything I set up, why I set it that way, and the things that are still open.

Files I left for you

File	Type	What it does
`run_trex_batch.sh`	bash	Runs TRex on every video in one folder, one after another.
`working.settings`	TRex parameters	All tracking settings, in one file the script reads.
`check_tracking_quality.py`	Python	Tells me how many frames the bee was actually found in, per video.
`bee_analysis.ipynb`	Jupyter	I open this when I want to look at one trajectory closely.

How to actually run it

1. Put all the videos in one folder, for example ~/Desktop/videos/.
2. Open run_trex_batch.sh in any text editor.
3. Change INPUT_DIR so it points to your video folder, and OUTPUT_DIR to where you want the results to land.
4. Change SETTINGS_FILE to the full path of working.settings.
5. In the terminal, switch on the conda environment with: conda activate tracking
6. Run the script
7. Now wait. Each video takes about 30 seconds to 2 minutes, depending on length.
8. When it's done, check the detection rates with: check_tracking_quality.py
9. To look at a single trajectory, open bee_analysis.ipynb in Jupyter, or something that you do better.

The settings file, line by line

The table below is everything inside working.settings. I grouped the lines by what I did to them compared to Fereshteh's original file.

Parameter	Value	Why this value
Kept from Fereshteh's original file		
calculate_posture	true	Asks TRex to also save the bee's body shape. We may need it later, so I left it on.
detect_type	background_subtraction	Catches the bee by spotting what moves against the still background. No ML model needed.
track_background_subtraction	true	Same idea, used during the tracking step (after the detection step).
track_max_individuals	1	Each of our videos has only one bee, so TRex should never look for more.
track_size_filter	[[0.08963, 0.89635]]	Drops anything too small or too big to be the bee. This is what filters the dust-like noise points.
detect_size_filter	[]	Empty on purpose. We rely on the track size filter above to do the filtering.
meta_encoding	gray	Reads the videos in grey. Faster, and the bee is dark on a light floor anyway.
meta_real_width	1280	Left as-is; the real calibration is set by cm_per_pixel below.
individual_image_scale	7	Default value. Not important here.
gui_show_video_background	false	Cleaner look in the TRex window. Has no effect on the data.
output_format	csv	I want CSV so I can open the files in Python, R, or even Excel.
Things I changed from her file		
cm_per_pixel	0.0612	Fereshteh had 0.0773 for her own camera. I measured the arena (82 cm) on a frame of my video and got 1340 pixels, so 82/1340 = 0.0612. Without this, all distances and speeds would be ~26% wrong. You can change it for different settings or you can check again this setting that might be different.

Parameter	Value	Why this value
`detect_threshold`	`7`	Was 10. I lowered it so the bee gets picked up more often, especially when it slows down near the feeder. On one test video, detection went from 2.6% to 6.4%.
`track_max_speed`	`30`	Was 24.7 cm/s. Small safety margin in case the bee makes a quick turn. Still in cm per second.
`video_conversion_range`	`[-1, -1]`	Was [0, 14796], which is the length of Fereshteh's one video. Our videos are different lengths (up to 6 min). -1, -1 just means "the whole video, whatever its length".
Things I removed from her file		
`gui_zoom_limit`	`(removed)`	Only remembers where her GUI was zoomed. Does nothing for tracking.
`manual_matches`	`(removed)`	A hand-made ID fix for one specific blob in her video. That blob doesn't exist in our videos, so it would only cause trouble.
`meta_source_path`	`(removed)`	The path to her video on her computer. The batch script gives the video with -i, so we don't need it.
`output_dir`	`(removed)`	Her own results folder. The script sets the output folder per video.
Things I added (new lines, not in her file)		
`track_max_reassign_time`	`120`	This was the most important one. Our bees sit still at the feeder for up to two minutes, and background subtraction loses them while they're still. Without this line, TRex would think a new bee appeared every time the bee started moving again. 120 means "if the bee comes back within 120 seconds, it's still the same one". After adding this, ID fragmentation went down to zero across 84 videos.
`track_trusted_probability`	`0.01`	Helps the same idea, accept the link even when TRex isn't very sure it's the same bee.
`match_min_probability`	`0.001`	Same again. Lets borderline matches pass instead of cutting the ID.

Things to watch out for

— Detection rates vary a lot between videos (1.5% to 100% in my test). Most of that is real: a bee that walks a lot is caught a lot. A bee that sits at the feeder isn't.

— Frames where the bee is still get marked as "missing". That makes pauses tricky to count.

— I measured the calibration on one video only. If different recording sessions used different camera positions, the calibration will be a bit wrong for some videos.